

Do Safety Monitors Fail Across Languages? — Validation & Execution Blueprint for Sana Ullah

TL;DR

Verdict: YELLOW (proceed, but only with a sharply narrowed contribution). The broad question "does CoT safety monitoring fail across languages?" was largely answered before Sana could start it — most decisively by Onyame et al., "The Fragility of Chain-of-Thought Monitoring Across Typologically Diverse Languages" (arXiv:2605.27901, 27 May 2026), which reports "an average rate of 95.9% across 8B–120B parameter models" and that "these deceptive patterns remain 100% in low-resource languages," and by Zhao et al. (Findings of EACL 2026, arXiv:2510.09555). A generic "we added Urdu" study is now RED. But three genuine, verifiable gaps survive: (1) **native-speaker validation** that automated multilingual monitors are actually trustworthy in a low-resource language, (2) **translate-the-CoT-then-monitor as a recovery mitigation**, and (3) **disentangling monitor/judge degradation from model unfaithfulness** — none of which the competing papers did.

The defensible contribution is a measurement-validity result, not a "more languages" result: demonstrate, with native Urdu ground truth, whether the field's standard automated CoT monitor is measuring genuine unfaithfulness or is itself failing in low-resource languages — and whether cheap post-hoc translation recovers the lost safety signal.

First supervisor to approach: Depeng Xu (UNC Charlotte, TAIMing AI center), who published "Fine-tuning LLMs with Cross-Attention-based Weight Decay for Bias Mitigation" (Haque, Fu, Xu, Yuan & Niu, Findings of EMNLP 2025, pp. 15785–15798). Start this week on free (Tier-A) resources by reproducing the Turpin/Chen English hint-faithfulness baseline on DeepSeek-R1-Distill-Qwen-7B via Colab.

Key Findings

1. The novelty landscape has moved fast and mostly against the original framing

The single most important fact for Sana: between the Run-1 recommendation and now (September 2026), the "monitorability across languages" space filled in. The nearest and most threatening paper is **Onyame, Zhou, Thopalli, Kailkhura & Agarwal, "The Fragility of Chain-of-Thought Monitoring Across Typologically Diverse Languages"** (University of Virginia + Lawrence Livermore National Laboratory; arXiv:2605.27901, v1 27 May 2026). It already delivers the headline result Sana's original idea would have claimed:

13 typologically diverse languages spanning high/mid/low-resource tiers, 16 models across 7 families (Qwen3, DeepSeek-R1 distills, GPT-OSS, Llama, Gemma 3, plus closed GPT/Claude).

Adversarial hint methodology (simple `<answer>C</answer>` cue + complex modular-arithmetic cue) on multilingual GPQA.

An automated LLM judge monitoring whether the CoT verbalizes the hint. (Note: the specific judge

model should be confirmed from the paper's Appendix D.2, "Judge Prompts for Deception Rate Evaluation"; a widely-cited attribution to "GPT-5.1" is *not* confirmed by public materials and Sana must verify it directly before citing.)

Headline numbers (verbatim from the abstract): "we consistently find CoT unfaithfulness across languages and hint types, with an average rate of 95.9% across 8B–120B parameter models"; "these deceptive patterns remain 100% in low-resource languages"; and "frontier models often commit to the misaligned cue in their latent activations within the first 15% of generation, even when the CoT appears faithful."

Their resource tiers are defined empirically: they prompt "Kimi K2.6... a strong open-source multilingual model, to solve the GPQA questions in a zero-shot setting," then tier languages by accuracy "above 60% for high-resource, between 30% and 60% for mid-resource, and below 30% for low-resource."

Combined with **Zhao et al. (EACL 2026 Findings)** — which already measured multilingual CoT *performance, consistency, and faithfulness* (including language compliance and truncation/error-injection faithfulness across languages including Bengali and Yoruba) — the plain statement "monitorability degrades in lower-resource languages" is **no longer novel**.

2. But the competitors left specific, defensible gaps — and they are exactly where Sana's native-speaker advantage matters

Verified against the Onyame et al. text and repository, the closest competitor did **not**:

(a) Use any native-speaker human validation. Its low-resource judgments rest entirely on an automated LLM judge, spot-checked only by the (non-native) authors. Their own words: "we use [the judge model] as the verification judge"; "We validate the judge through manual inspection of samples across all languages." No bilingual/native annotator is involved. [arxiv](#)

(b) Test translate-the-CoT-back-to-English-then-monitor as a recovery mitigation. Their stress tests cover option-letter bias, closed-source generalization, and stochastic stability only. [arxiv](#)

(c) Separately quantify whether the monitor/judge itself degrades in low-resource languages, as distinct from the model's own unfaithfulness. A monolingual-English-optimized judge reading Urdu CoT could be the thing failing — a confound that directly threatens the "100% in low-resource" claim.

(d) Include Urdu among its named low-resource languages (its low-resource exemplars are Swahili, Telugu, Yoruba). The full 13-language list is given in the paper's "Table 6... grouped into 8 typologically diverse language families" (Appendix C), so Sana can and must confirm Urdu's presence/absence directly there before asserting it.

It **did** partially control for base accuracy (a "Baseline Error Rate" analysis in Appendices B.1/B.6 showing deception persists even when accuracy is comparable) — so Sana must **not** claim competitors ignored task difficulty. The residual gap is judge-side degradation, not accuracy control.

Crucially, no 2025–2026 paper combines **native-speaker low-resource CoT-monitorability validation**

with a **post-hoc CoT-translation recovery method**. That intersection is genuinely open. The nearest native-validation precedent is a Persian-language study ("A Comprehensive Evaluation of Chain-of-Thought Faithfulness in Persian Classification Tasks," LoResLM 2026), which used native human evaluation of faithfulness but did *not* address monitorability, Urdu, or translation recovery.

3. Is the refined title stronger than Run 1's? Yes — but it still needs to be narrowed further

"Do Safety Monitors Fail Across Languages?" is a better *frame* than "Cross-Lingual Faithfulness and Monitorability" because it names a falsifiable safety claim. But as a *project scope* it is now too broad and collides head-on with Onyame et al. The winning move is to **reframe from "do monitors fail?" (answered) to "when monitors appear to fail in a low-resource language, is that real model unfaithfulness or measurement artifact — and can we fix it?" (open)**. This converts a crowded claim into a measurement-validity + mitigation contribution where native Urdu competence is a genuine methodological asset rather than a cosmetic add-on.

4. Faculty reality check (verified September 2026)

Depeng Xu — Assistant Professor, Software & Information Systems + School of Data Science; core member of UNC Charlotte's Center for Trustworthy AI through Model Risk Management (TAIMing AI). Published "Fine-tuning LLMs with Cross-Attention-based Weight Decay for Bias Mitigation" (with Farsheed Haque, Zhe Fu, Shuhan Yuan, Xi Niu) at **Findings of EMNLP 2025, pp. 15785-15798**; the work "received a 2025 TAIMing AI Seed Grant Award." Research: trustworthy AI, algorithmic fairness, robust/explainable ML in text. **Best current fit and #1 target.**

Razvan Bunescu — Associate Professor, CS; active NLP/ML researcher with 2025 work on LLM *reasoning trajectories* (Socratic debugging; EMNLP 2025 recommender paper). Strong reasoning-adjacent fit; #2.

Wlodek Zadrozny — Professor, NLP; **transitioning to Emeritus status in 2026** — availability risk, deprioritize as primary.

Samira Shaikh — retains a UNC Charlotte Cognitive Science/CS appointment but is now **Director of Data Science at Ally (industry)**; no verifiable 2025-2026 LLM-faithfulness publications (most recent indexed NLP paper is LREC-COLING 2024). Interesting historical Urdu-corpus work but **not currently an active supervisor for this topic.**

Liyue Fan — trustworthy/privacy AI; adjacent, not reasoning-focused.

5. Harvard alignment is real but should be framed honestly

The single strongest intellectual match is **Himabindu Lakkaraju** (Harvard; her group produced Xiong, Chen, Qi & Lakkaraju, "Measuring the Faithfulness of Thinking Drafts in Large Reasoning Models," NeurIPS 2025 — the counterfactual draft-perturbation method Sana should adopt). Alignment is **strong** on trustworthy-AI/fidelity, **moderate** on the multilingual angle. No public evidence about current admissions should be treated as a promise.

Details

PHASE 0 — Adversarial novelty check and comparison table					
The comparison table (all papers verified real, with arXiv/ACL identifiers):					
Paper	Year/Venue	Languages	Models	Faithfulness method	Safety/focus
Onyame et al. (arXiv:2605.27901)	2026 preprint (UVA+LLNL)	13 typologically diverse (Swahili, Telugu, Yoruba as low-res)	16 (Qwen3, R1-distills, GPT-OSS, Llama, Gemma3, GPT, Claude)	Adversarial hints (simple+complex) + logit lens	Yes (comprehensive)
Zhao et al. (arXiv:2510.09555)	Findings EACL 2026	Multiple incl. Bengali, Yoruba	R1-Qwen-32B, R1-Llama-70B, etc.	Truncation + error injection; consistency; language compliance	Partial
Qi et al. (arXiv:2505.22888)	Findings EMNLP 2025	Multiple	2 LRM families (XReasoning)	Language-compliance + accuracy tradeoff	Oversight framing
Xiong, Chen, Qi & Lakkaraju (arXiv:2505.13774)	NeurIPS 2025 (Harvard)	English	6 LRMs	Counterfactual draft perturbation	Monitor motivational
Chen et al. (arXiv:2505.05410)	2025 (Anthropic)	English	Claude 3.7, R1	Hint-verbalization reveal rate	Yes
Turpin et al. (arXiv:2305.04388)	NeurIPS 2023	English	GPT-3.5, Claude 1	Biasing features (answer-always-A)	Foundational
Lanham et al. (arXiv:2307.13702)	2023 (Anthropic)	English	Claude family	Perturb CoT (mistakes/paraphrase/truncation)	Foundational
Emmons et al.	2025 (Google)	English	Gemini	CoT-as-computation vs	Yes

(arXiv:2507.05246) DeepMind)			family	rationalization	
Korbak et al. (arXiv:2507.11473)	2025 (multi-org)	English	—	Position paper	Yes (def monitor)
Yang et al. (arXiv:2511.08525)	2025→2026	English	LRMs	Verbalization + monitor reliability	Yes
"Persian CoT Faithfulness" (aclanthology 2026.loreslm-1.27)	2026 LoResLM	English, Persian	6 LMs	LLM-as-judge + human eval	Faithful
UrduBench (arXiv:2601.21000)	2026	Urdu	Reasoning + instruct LLMs	Accuracy + language consistency (not faithfulness)	No

Verdict: YELLOW. Proceed only with the narrowed measurement-validity + mitigation contribution. The original "add Urdu to a monitorability study" is RED. The Persian LoResLM 2026 paper is instructive: it proves native-language faithfulness validation is publishable and *not yet done for monitorability or for Urdu*, and it did not test the translate-then-monitor recovery.

PHASE 1 — Operationalizing "monitorability" as measurable variables

Adapting Turpin 2023, Lanham 2023, Chen 2025, Xiong/Lakkaraju 2025, Zhao 2026, Korbak 2025:

Faithfulness (verbalization/disclosure): disclosure rate = $P(\text{CoT explicitly mentions dependence on cue } h \mid \text{model reached the hinted answer})$.

Intervention sensitivity (Lanham): Δ in answer distribution when the CoT is truncated/corrupted/paraphrased. High sensitivity $\Rightarrow$ CoT is load-bearing (CoT-as-computation).

Answer-switch rate (Turpin/Chen): $P(\text{answer changes from no-hint baseline to hinted answer})$ — the *influence* signal.

Monitor detection rate / monitor AUROC: the monitor's ability, given the CoT, to correctly flag "this trace was influenced by a cue," scored against ground-truth influence labels; report precision/recall/F1 and AUROC.

Hidden influence: answer-switch = yes but disclosure = no — the safety-critical quantity.

Language compliance (Zhao/Qi): fraction of the CoT actually generated in the requested language (measured, not assumed).

Reasoning consistency / answer consistency: cross-language agreement of reasoning content and of final answers.

Calibration: correspondence between stated confidence and correctness — secondary, only if time

allows.

Monitorability (composite, Korbak/Emmons): degree to which reading the CoT lets an external monitor identify safety-relevant influence $\approx$ disclosure rate $\times$ monitor detection rate $\times$ language compliance.

Monitor-validity gap (Sana's novel measured quantity): (native-human monitor detection) – (automated monitor detection), per language.

PHASE 2 — Research questions and falsifiable hypotheses

Primary RQ: When an automated CoT safety monitor reports degraded monitorability in a low-resource language (Urdu), how much is *genuine model unfaithfulness* versus *monitor/judge measurement failure* — and does translating the CoT to English before monitoring recover the lost safety signal?

H1 (model effect): Hidden-influence rate is higher in Urdu than English even under native-human monitoring. *Null still publishable:* if native humans detect influence equally well across languages, the "low-resource fragility" claim is partly a measurement artifact.

H2 (monitor-validity gap): The automated English-centric monitor's detection rate drops more than native-human detection in Urdu ($\text{gap} > 0$). *Falsifiable:* $\text{gap} \approx 0 \Rightarrow$ automated monitors are trustworthy in Urdu.

H3 (recovery): Translating the Urdu CoT to English before monitoring raises automated detection toward the English baseline. *Falsifiable:* no recovery $\Rightarrow$ safety-relevant information is lost at generation, not at monitoring.

H4 (dose-response): Monitor-validity gap increases monotonically with decreasing resource level after controlling for base accuracy. *Null:* effect vanishes after accuracy control $\Rightarrow$ monitorability tracks competence, not language per se (kill trigger D).

H5 (compliance mediation): Degraded monitorability is mediated by low language compliance (mid-trace reversion to English).

PHASE 3 — The unique, peer-review-defensible contribution

A native-validated measurement-validity study of cross-lingual CoT monitorability, plus a translate-then-monitor recovery test. Three deliverables:

Native Urdu ground-truth for whether traces are genuinely unfaithful — the first such validation for monitorability in Urdu, isolating the *monitor-validity gap* the competing papers could not measure.

Evidence on whether post-hoc CoT translation recovers safety signal — a cheap, deployable mitigation nobody has tested.

A reproducible protocol/benchmark separating three often-conflated quantities: answer correctness, CoT faithfulness, and monitor detectability. Explicitly *not* "Zhao/Onyame + Urdu."

PHASE 4 — Principled language ladder

English (en) — high-resource anchor, Latin script.

German (de) or Spanish (es) — high-resource control. Latin script (isolates "not-English" from "low-

German (de), or Spanish (es) — high-resource control; Latin script (isolates *script* from *resource level* vs. Urdu).
resource").

Arabic (ar) — mid/high-resource, **Perso-Arabic script** (isolates *script* from *resource level* vs. Urdu).

Hindi (hi) — Hindustani sibling, **Devanagari script**, higher resource than Urdu (isolates *script* from *linguistic content*).

Urdu (ur) — target low-resource, Perso-Arabic script, native validation available.

Bengali (bn) or Yoruba (yo) — very-low-resource lower anchor (matches competitors' tiers).

Operationalizing "resource level": composite of (i) estimated pretraining token share, (ii) benchmark availability, (iii) **tokenizer fertility** (tokens-per-word, a computable representation proxy), (iv) base-model accuracy (this is exactly how Onyame et al. tier languages — via zero-shot GPQA accuracy on a strong multilingual model), (v) linguistic distance from English, (vi) script. **Confound separation**: the Hindi-Urdu pair holds language ~constant while varying script/resource; the Arabic-Urdu pair holds script family ~constant while varying language; German/Spanish holds resource high while varying "Englishness." Translation artifacts controlled by native Urdu correction (Phases 6/9); difficulty controlled by item-matched subsets and base-accuracy covariates.

PHASE 5 — Models (open-weight, inspectable traces, feasible Sept 2026)

DeepSeek-R1-Distill-Qwen-7B — 7B; strong math reasoning; Qwen multilingual base; open; full

`<think>` trace; single 16GB GPU / Colab. **Primary Tier-A workhorse and English-baseline replication target** (Onyame et al. use the DeepSeek-Qwen distills, enabling direct comparison).

DeepSeek-R1-Distill-Qwen-14B — 14B; UrduBench found it sustains difficulty well; needs ~24-40GB or quantization. **Tier-B/C main model.**

Qwen3-8B (thinking mode) — 8B; **~119-language** coverage (vs. ~30 for the DeepSeek distills) — best multilingual reasoner in the accessible class; open; single-GPU. **Primary for the multilingual arm.**

Qwen3-32B — stronger; Tier-C.

DeepSeek-R1-Distill-Llama-8B — different family for generalization; open; single-GPU.

GPT-OSS-20B — open-weight, different family/scale; Tier-B/C for family generalization.

Monitor/judge models: an open judge (e.g., Qwen3-32B or a Llama-70B-class model), plus optionally one API judge for calibration — but the *scientific* comparison is **automated judge vs. native human**.

Avoid designing around closed models (GPT-5.x, Claude) whose full traces aren't inspectable, except optionally as an extra external judge.

PHASE 6 — Datasets/tasks

MGSM (math, multi-step) — 250 items, 10+ languages incl. Bengali; already Urdu-translated in

UrduBench. Clean for hint injection. Contamination: moderate — mitigate with paraphrase checks.

GPQA / GPQA-Diamond (hard science MCQ) — the competitors' dataset; enables direct comparison; needs Urdu translation + native correction; low contamination.

CommonSenseQA / OpenBookQA (commonsense/factual) — in UrduBench; MCQ ideal for cue injection.

BBH / BBQ-style items (bias/hint-sensitive) — for the biasing-feature intervention; BBQ needs cultural adaptation.

A small safety-relevant decision set — ~50 items where a cue nudges toward an unsafe/wrong recommendation (ethically bounded, no operational-harm content).

Native Urdu subset construction (required): ~150–250 MCQ items stratified across math/commonsense/science. Pipeline: (1) machine-translate; (2) **native-speaker (Sana) correction** for fluency/terminology/answer-preservation; (3) **back-translation check** + second-native-reviewer adjudication on a 50-item audit; (4) validation criterion $\geq 90\%$ native agreement that meaning and correct answer are preserved — benchmarked against the documented ~60% raw-MT label agreement for Urdu (vs. ~90% English), which is precisely why native correction is scientifically necessary. Target $n \approx 200$ items $\times$ ~5 seeds per condition.

PHASE 7 — Faithfulness protocols compared

Turpin (biasing features): reorder options / suggested answer. Influence = accuracy drop; disclosure = CoT mentions bias. Simple, strong effect; but bias may be too easy (Emmons critique); option-letter bias interacts with script/RTL.

Lanham (CoT perturbation): truncate/insert-mistake/paraphrase, measure answer change. Measures load-bearingness; needs controlled decoding; truncation points differ by tokenization.

Chen 2025 (hint reveal rate): six hint types; the current standard, directly comparable; "used the hint" is judge-inferred.

Xiong/Lakkaraju 2025 (counterfactual draft perturbation): intra-draft + draft-to-answer; causal, fine-grained; heavier engineering; English-only so far.

Zhao 2026 (multilingual truncation + error injection): already cross-lingual and reusable; no native validation, no monitor-validity measurement. **Recommendation:** reproduce the **Turpin/Chen hint-based method first** (simplest, matches Onyame's setup for direct comparison), then layer Zhao's multilingual truncation as a robustness check.

PHASE 8 — Centerpiece monitorability experiment

Design: hidden influence $\rightarrow$ model reasoning $\rightarrow$ answer; then CoT $\rightarrow$ independent monitor $\rightarrow$ detect influence?; across the ladder; with the **native-human monitor as ground truth**.

Hypothesis: H2 (gap > 0 in Urdu) + H3 (translation recovers).

IVs: language (ladder); cue type (incorrect-answer suggestion, biasing evidence, shortcut feature); monitor type (automated-English-judge / automated-in-language-judge / **native human** / translate-then-English-judge).

DVs: answer-switch rate, disclosure rate, hidden-influence rate, monitor detection rate/AUROC, language compliance.

Controls: no-hint baseline; base-accuracy covariate; option-letter-bias control; ≥ 5 seeds.

Sample size: ~200 items $\times$ 6 languages $\times$ 3 cue types $\times$ 5 seeds (power to detect a 10-point detection-rate gap at $\alpha=.05$, power .8 — smaller than the tens-of-points effects typical in this literature).

Test: mixed-effects logistic regression (item + model random effects), language fixed, base accuracy covariate; AUROC via DeLong.

Expected result: automated detection drops in Urdu; native detection drops far less; translation

partially recovers automated detection.

Interpretation: native $\approx$ automated $\rightarrow$ validates competitors' automated-only method (useful confirmation); native $\gg$ automated $\rightarrow$ the "100% low-resource deception" literature partly reflects *monitor* failure (high-impact corrective).

PHASE 9 — Control experiments

Translation quality (native validation); task-difficulty matching (item-matched subsets + covariate); base-accuracy control (analyze only items solved without hint, or covary accuracy); **language compliance** (measure % Urdu, flag reverts); **script vs. language** (Hindi/Arabic/Urdu triad); prompt-template sensitivity (≥ 3 templates); model-family generalization (Qwen + Llama + GPT-OSS); model-size scaling (7B/14B/32B); repeated generations (≥ 5 seeds, temperature reported); contamination/memorization checks (paraphrase items).

PHASE 10 — Statistical analysis plan

Metrics: disclosure rate, hidden-influence rate, monitor AUROC + P/R/F1, answer-switch rate, language-compliance rate, monitor-validity gap (native – automated), all with bootstrap 95% CIs. Tests: mixed-effects logistic regression; DeLong for AUROC; McNemar for paired human-vs-automated on identical traces; Cohen's κ for native inter-annotator agreement. Multiple comparisons: Benjamini-Hochberg FDR across language $\times$ cue cells. Meaningful effect size: given disclosure effects often $< 20\%$ and Urdu vs. English label-agreement gaps of ~ 30 points, treat a **≥ 10 -point monitor-validity gap** as scientifically meaningful.

PHASE 11 — Mitigation

Primary: **translate-the-CoT-to-English-then-monitor** (verified *not* done by competitors). Secondary (only if focus permits): in-language vs. English monitor; ensemble monitoring; language-aware calibration thresholds; prompting the model to reason in-language (Qi et al. show this aids oversight but costs accuracy — cite the tradeoff). Pursue mitigation only after the core measurement result is locked.

PHASE 12 — Repository structure



xliling-monitor/
data/ # raw, translated, native-corrected, splits (DVC or git-lfs)
src/ # generation, hint-injection, monitors, translation
models/ # model configs, loaders (vLLM)
experiments/ # one config-driven script per experiment
configs/ # YAML: model, language, cue, seed
notebooks/ # exploratory only
evaluation/ # metrics: disclosure, AUROC, compliance
statistics/ # mixed-effects, DeLong, FDR
results/ # versioned JSONL outputs
figures/ paper/ tests/

Environment: `uv`/`conda` + pinned `requirements.txt`; vLLM for inference; global seed control; Hydra/OmegaConf config management; structured JSONL logging (model, lang, seed, prompt hash); W&B or CSV logging; deterministic decoding option; a `make reproduce` target. This leverages Sana's backend/data-pipeline strengths directly.

PHASE 13 — Compute & budget

Tier A (free — start now): laptop + Colab free (T4 16GB). DeepSeek-R1-Distill-Qwen-7B / Qwen3-8B via vLLM or 4-bit. English baseline + Urdu pilot on ~50 items. GPU-hours: tens. Cost: \$0. Storage <20GB.

Tier B (Colab Pro / rented A100 40GB, ~\$1-2/hr): full ladder × 7B/14B × 5 seeds. ~100–300 GPU-hours; ~\$150–600. API judge (optional): tens of dollars. Storage ~100GB.

Tier C (university lab GPUs, via TAIMing AI / advisor): 32B models, full replication, family generalization. ~500–1000 GPU-hours; marginal cash cost.

This week on Tier A: reproduce Turpin/Chen English hint-faithfulness on 7B; stand up the repo; translate+native-correct a 50-item Urdu pilot.

PHASE 14 — UNC Charlotte supervisors (verified) + outreach

Ranked: **(1) Depeng Xu** — trustworthy AI, LLM bias, TAIMing AI center, EMNLP 2025 + seed grant; direct fit + likely GPU access + grant infrastructure. **(2) Razvan Bunescu** — NLP/reasoning, active 2025 LLM reasoning-trajectory work; strong methodological mentor. **(3) Liyue Fan** — trustworthy/privacy AI; adjacent. **(4) Wlodek Zadrozny** — NLP but Emeritus-transition 2026 (availability risk). **(5) Samira Shaikh** — Urdu history but now industry, thin recent output; only as informal advisor.

Outreach email to Xu (#1), framed as evidence not need:

Subject: Undergraduate research — validating cross-lingual CoT safety monitors (reproduced a baseline)

Dear Dr. Xu, I'm a CS undergraduate (GPA 3.78, Class of '27) and a native Urdu speaker. I've been

studying chain-of-thought safety monitoring and noticed that recent multilingual results (Onyame et al. 2026; Zhao et al., EACL 2026) rely entirely on automated, English-centric LLM judges — with no native-speaker validation and no test of whether translating the reasoning trace back to English recovers the safety signal. I reproduced the English hint-faithfulness baseline (Turpin/Chen) on DeepSeek-R1-Distill-Qwen-7B and ran a small Urdu pilot; my early results suggest the automated monitor may itself be failing in Urdu, rather than the model being more deceptive. I'd value 20 minutes of your guidance on whether this "monitor-validity gap" framing is sound and worth developing, given your work on trustworthy LLM evaluation (e.g., your Findings of EMNLP 2025 bias-mitigation paper). A one-page design and repo are attached.

PHASE 15 — Harvard alignment (honest)

Himabindu Lakkaraju — strong on faithfulness/trustworthy AI (thinking-draft faithfulness, NeurIPS 2025); moderate on multilingual. Her frequent collaborators (Chen, Qi, Xiong) work directly on CoT faithfulness/monitorability — the exact neighborhood of this project. Treat as research alignment only; make **no admissions claim**.

PHASE 16 — Publication strategy

Realistic: ACL/EMNLP/NAACL/EACL **Findings**, or a workshop with proceedings — *LoResLM* (the Persian-faithfulness venue), *BlackboxNLP*, *TrustNLP*, or a NeurIPS/ICLR safety/SoLaR workshop.

Stretch: main-conference short paper at ACL/EMNLP, or a FAccT paper if the oversight/fairness framing is strong.

Fallback: arXiv preprint + non-archival workshop talk; still valuable for the letter. Avoid predatory venues. Likely deadlines: ACL cycle (Feb), EMNLP (May), NeurIPS workshops (summer), EACL/LoResLM (fall). Contribution needed: workshop = one clean native-validated result; Findings = full ladder + mitigation.

PHASE 17 — Paper outline (evidence each section must eventually hold)

Intro (the monitor-validity problem); Related Work

(Turpin/Lanham/Chen/Emmons/Korbak/Zhao/Onyame — position explicitly); Definitions (Phase 1 variables); Cross-Lingual Faithfulness Protocol (hint method + multilingual adaptation); Monitorability Framework (native vs automated vs translate-then-monitor); Experimental Setup (models, ladder, datasets, native subset); Results (disclosure/hidden-influence/AUROC by language + monitor-validity gap); Analysis (accuracy control, script vs language, compliance mediation); Mitigation (translation recovery); Limitations; Ethics (safe cues, dual-use, native-annotator credit); Conclusion. No fabricated numbers — populate only after experiments.

PHASE 18 — Kill/pivot criteria

(A) A new paper does essentially this exact study (native Urdu validation + translate-then-monitor). → Pivot to a *different* mitigation (in-language monitors/ensembles) or a different low-resource language pair.

(B) English baseline faithfulness effect won't reproduce. → Debug harness; if truly absent, pivot to

(B) English baseline faithfulness effect won't reproduce. → Debug harness, if truly absent, pivot to a multilingual language-compliance/consistency study.

(C) Models can't reliably reason in Urdu (compliance too low). → Pivot to the *compliance-as-oversight-bottleneck* angle (builds on Qi et al.).

(D) Language differences vanish after base-accuracy control. → Report the null as a corrective (still publishable); shift emphasis to measurement methodology.

(E) The metric mostly measures translation quality. → That *is* the finding; reframe the paper around measurement validity.

PHASE 19 — Roadmap

Weeks 1-8: W1 literature mastery + reproduce English hint baseline on 7B; W2 repo + config harness + metric unit tests; W3 Urdu 50-item pilot (translate+native-correct); W4 run pilot ladder (en/hi/ur), inspect monitor-validity gap; W5 email Xu with pilot evidence; W6 expand to full native subset (~200 items) + add Arabic/Bengali; W7 add second model family + seeds; W8 first-pass statistics + go/no-go against kill criteria. **Months 3-6:** full ladder × models × cues; native inter-annotator study; translate-then-monitor mitigation; secure supervision + lab GPU (Tier C). **Months 6-9:** analysis, robustness/controls, draft workshop paper; post preprint. **Months 9-12:** submit workshop/Findings; iterate. **Months 12-18:** camera-ready/second study; recommendation-letter timing aligned to fall PhD deadlines; PhD applications with preprint + letter in hand.

PHASE 20 — First 72 hours

Day 1 (read): Onyame et al. 2605.27901 (§3 setup, §7 stress tests, App. B-C incl. Table 6 to confirm whether Urdu is included, App. D.2 to confirm the judge model); Chen et al. 2505.05410 (hint method); Turpin 2305.04388 (method); Zhao 2510.09555 (multilingual faithfulness metrics). Write a one-paragraph statement of the monitor-validity gap.

Day 2 (build): create the repo per Phase 12; `pip install vllm transformers`; load DeepSeek-R1-Distill-Qwen-7B on Colab; implement `inject_hint(item, cue)` and a keyword+LLM disclosure checker; unit-test on 5 items.

Day 3 (baseline target): reproduce the English hint effect — measure answer-switch and disclosure rate on ~50 MGSM items with/without the incorrect-answer cue; **target: observe answer-switching with disclosure well below switching (the Turpin/Chen signature; Chen et al. report reveal rates "often below 20%").** If reproduced, translate 10 items to Urdu (native-correct) and check whether the automated judge still reads them — the first glimpse of the gap.

Recommendations

Reframe now, before coding. Keep the title if you like, but the *contribution* must be "native-validated monitor-validity + translation recovery," not "monitorability across languages." Without this narrowing the project is RED against Onyame et al.

This week (Tier A, free): reproduce the English hint baseline on 7B and run a 10-50 item Urdu pilot. Evidence of a monitor-validity gap is your single most valuable asset for both faculty outreach and venue selection.

Approach Depeng Xu first with the evidence-based email; Bunesco as strong backup. Do not lead with Harvard.

Lock the confound controls early (base-accuracy covariate, Hindi/Arabic/Urdu script triad, language-compliance measurement). These separate a publishable result from a rejected "we added Urdu" paper.

Benchmarks that change the plan: native-vs-automated gap <10 points → pivot to mitigation/compliance framing (kill D/E); a competing native+translation paper appears → pivot language pair or mitigation (kill A).

Direct answers to the ten required questions: (1) Yes, the refined framing is stronger than Run 1, but only after further narrowing to measurement validity. (2) Yes, a defensible window remains after Zhao 2026 and Onyame 2026 — in native validation, translation recovery, and judge-vs-model disentanglement. (3) The novel contribution is native-Urdu-validated evidence on whether automated cross-lingual CoT monitors measure real unfaithfulness or fail themselves, plus a translate-then-monitor recovery test. (4) The convincing experiment is the four-monitor comparison (automated-English / automated-in-language / native-human / translate-then-English) across the language ladder with base-accuracy controls. (5) The result that would genuinely interest the community: native humans catch influence that the automated monitor misses in Urdu (monitor-validity gap ≥ 10 points), and cheap translation recovers most of it — a corrective to the "100% low-resource deception" narrative. (6) Depeng Xu first. (7) The Turpin/Chen English baseline on DeepSeek-R1-Distill-Qwen-7B plus a native Urdu pilot, on free Colab, this week. (8) Any of kill criteria A–E, each with a mapped pivot. (9) Workshop-with-proceedings (LoResLM/TrustNLP/BlackboxNLP) or a *CL Findings paper, with arXiv preprint as fallback. (10) The complete blueprint above (Phases 1–20).

Caveats

The specific automated-judge model used by Onyame et al. and the full 13-language list are stated in the paper's appendices (Table 6, App. C; judge in App. D.2) but were not confirmable from public secondary materials; a common "GPT-5.1 judge" attribution is **unverified** and Urdu's absence is **likely but not confirmed** — Sana must check both directly in the PDF before citing.

Many cited works (including the competitor and several 2026 titles) are **arXiv preprints, some not yet peer-reviewed**; numbers may change on revision, and future-dated IDs should be re-verified at project start.

No statement here implies any Harvard (or other) admission outcome; PhD admission is not predictable from a single project.

Ethical scope: use only benign, controlled cues; do not construct operational-harm content; credit native annotators (including Sana) explicitly in any publication.